

VUKA — you don't have to ask.

An Android safety app that notices duress without a button, alerts a guardian silently, and keeps a record anyone can check on a public ledger. *Vuka* is isiZulu for “wake up”.

Member	University	Role
Lethabo Hoaeane	UNISA	Co-lead: architecture, product, UX
Sibusiso Khumalo	Wits	Co-lead: backend, ledger, CI, Azure deployment
Babatunde Adelusi	University of Pretoria	Business case, economics, pitch
Mutarisi Chibaya	University of Pretoria	Android app and guardian screens
Khutso Mothopa	Wits	Requirements, evidence, traceability
Vukosi Khoza	Wits	On-device sensing, signing, measurement
Ipeleng Constance Modise	TUT	Threat model, secure lifecycle, privacy

Try it and check it	
Code (public)	https://github.com/LethaboMH14/V.U.KA--Geekulcha
Public ledger (verify a record)	https://lethabomh14.github.io/V.U.KA--Geekulcha/dashboard/ledger/
Android app — VIGIL 0.0.15 (APK)	https://github.com/LethaboMH14/V.U.KA--Geekulcha/releases/download/vigil-demo/VIGIL.apk
Demo video	Uploaded with this submission (46 s)

How to read this deck. Every figure is tagged: **FACT** (published or measured), **ESTIMATE** (derived by our script), **ASSUMPTION** (believed, not sourced), **PROPOSED** (designed, not built). Each slide names the judging criteria it answers: I innovation · T technical · U usability & design · S security & ethics · B business · Q quantum bonus.

One evening, and no proof.

Illustrative composite built from the cases on the next slides — not a real person.

Friday, 19:00. Thandi gets into her car outside a mall in Pretoria. Two men get in with her. They hold her phone to her face; it opens. She types the amount they tell her, then they take her to an ATM for the daily limit.

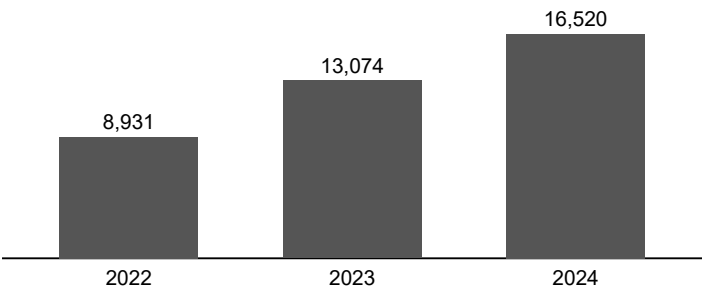
On Monday the bank shows her the transaction: her phone, her face, her PIN. Approved. Her insurer asks for proof she was forced. She has none.

The gap in one line: a coerced transfer authenticates exactly like a voluntary one, and no party holds a verifiable record of the coercion.

What happens today	Source
Insurers verify coercion after the fact: bank statements, ATM footage, a sworn affidavit	iTOO via FAnews, 19 Sep 2024 · FACT
The ombud held that a victim who surrendered credentials under duress had no grounds to hold the bank liable	NFO via FAnews, 19 Sep 2024 · FACT
Banks are not obliged to reverse transactions on request	OBS Annual Report 2022 · FACT
Every safety tool needs a free hand: a panic button must be pressed; a bank duress code must be typed, inside one bank	FNB GuardMe; Namola; Discovery Panic Code (MyBroadband, 27 Mar 2025) · FACT

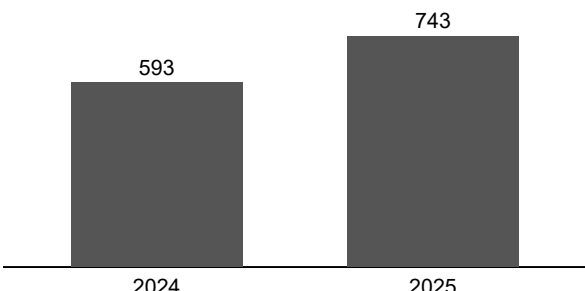
Fraud pressure and disputes are rising — and coercion is invisible in every number.

Fraud & dishonesty cases detected (ASISA members)



ASISA media releases · FACT (not yet in docs/EVIDENCE.md)

Banking formal cases per month at the ombud (NFO)



NFO Annual Report 2025 · FACT · +25.4%

Figure	Value	Source · tag
Digital banking crime, 2025	R2.4bn, 110,074 incidents — all digital banking crime, mostly social engineering, not our market size	SABRIC 2025 · FACT
Kidnappings, 2023/24	17,061 — up 264% in a decade; 44% during a hijacking	SAPS via ISS Africa, 3 Dec 2024 · FACT
Cost of one formal ombud case	R4,962.38	OBS Annual Report 2024 · FACT
Fraud complaint resolution	~61 days on average	NFO via The Citizen, 16 Mar 2026 · FACT
Banking determinations for the complainant	17%	NFO AR via Moonstone, 16 Jul 2026 · FACT
Mandatory reimbursement in South Africa	None	World Payments Monitor, 8 Aug 2026 · FACT

Reading it honestly: none of these counts coerced transfers, because a coerced transfer looks like a voluntary one. Its size is unknown and bounded above by the kidnapping figures; a pilot must measure it (next slide).

We do not invent the size of the problem.

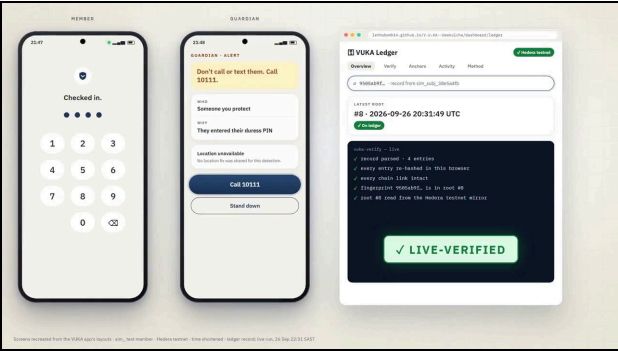
Unknown	What the pilot measures
Coerced transfers per year (no one publishes it)	Count from the partner insurer's or bank's own book
Insurer's cost per coerced-transfer dispute	Time-and-cost study of the partner's claim files
A bank's own cost per escalated dispute (R4,962 is the ombud's system cost)	The bank's own case cost
Frequency × severity of coerced-transfer claims	The actuary test: the insurer must save ≥ R240 per member per year (12 × R20)
Whether South Africa adopts mandatory reimbursement	Monitor FSCA, SARB and National Treasury

Regulatory direction	Position
United Kingdom	Mandatory push-payment reimbursement since 7 Oct 2024, capped at £85,000 (UK PSR) · FACT
South Africa	No mandatory reimbursement; losses generally borne by the consumer · FACT
National Treasury, Jun 2026	Aware of the surge; any response “would need to be carefully assessed” · FACT
SARB	Draft Fraud Risk Framework; Confirmation of Payee work · FACT

So what: today the bank's cost is dispute handling and churn, and the question of who carries the loss is open and moving.

VUKA notices duress without a button — and proves when it happened.

Feature	What it does	Status (27 Sep)
Always on after set-up	VIGIL listens on the phone with an on-device sound model (YAMNet); audio never leaves the phone	Built · VIGIL 0.0.15
Graded detection	Scream, shout, breaking glass, gunshot; clear sounds ask; weaker ones are recorded unless other signs back them up	Built · thresholds UNCALIBRATED
Journey check	“Are you okay?” opens over the lock screen; the normal PIN raises no alert	Built · VIGIL 0.0.15
Duress PIN	Same “Checked in” screen; guardians are alerted silently	Built · server rule: duress PIN → alarm
Wrong PIN	Guardians get a heads-up — never treated as duress, never sent to the bank; the screen still says “Try again”	Built · live on Azure 27 Sep (5/5 test run)
No answer	Guardians are alerted after the check-in window (about 70 s)	Built · timings PROPOSED
Hold for help	Hold 2 s, or use the Quick Settings tile: the same check-in a sound would open	Built · VIGIL 0.0.15
Guardian alert	Who and why in words; a map of the phone's location during the alert; Open in Maps; Call 10111	Built · background alerts; push (FCM) not configured
Signed, anchored record	Each event signed on the phone (P-256, Android Keystore), hash-chained on the server; a 33-byte Merkle root goes to Hedera testnet	Live · root #8 LIVE-VERIFIED
Anyone can verify	Paste a record into the public ledger; the browser recomputes it and reads the root from Hedera	Live · 26 Sep, test member
Sign-up with email code	The server sends real codes (first delivered 27 Sep 04:28 SAST, n = 1, from the native build); VIGIL 0.0.15's code step is still simulated	Server live · app step simulated
Post-quantum attestation	Hybrid ML-DSA-65 + Ed25519 over an anchored root	Demonstration · PR #117



From the demo video: the member's check-in, the guardian's alert, and the record verified on the public ledger. Screens recreated from the app's layouts; test member (sim_); Hedera testnet; time shortened.

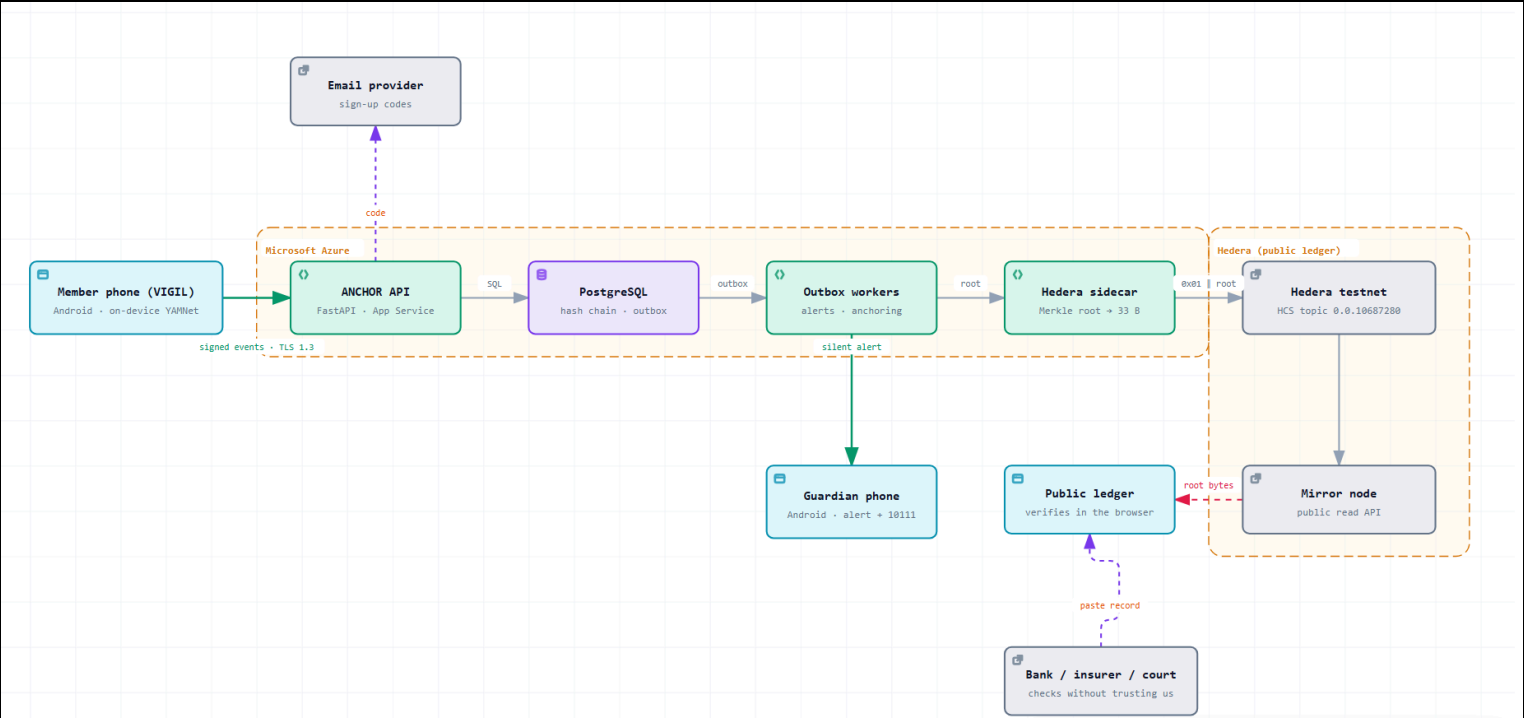
Everyday noise vs real danger: a sound alone never alerts anyone. It opens a check-in; the person answers — or the silence does. The machine never decides what happens to a person.

From a sound, to a person, to a proof.

When	Thandi (member)	VUKA	Her brother (guardian)	Later: bank or insurer
Before	Sets up her profile, a normal and a duress PIN; invites her brother	Registers her phone's key; issues a one-time invite code; shows "Guardian added" once he accepts	Enters the code, agrees, is linked	—
19:00	Locks the phone and puts it in her bag	Listening on the phone (always on); a notification shows it is on	Standby	—
19:07	Glass breaks; a shout	Hears "Glass" on the phone; opens a Journey check over the lock screen	—	—
19:07	Watched, she enters her duress PIN	Shows the same "Checked in"; sends a signed check-in result	—	—
19:08	Nothing visible changes	Opens an incident, alerts the guardian and shares the phone's location while the alert is open	Sees why in words and a map; taps Call 10111 (opens the dialer)	—
After	Safe	Every step is in her chained record; a 32-byte root is anchored on Hedera	Stands down; his acknowledgement is signed	Checks her record on the public ledger without trusting VUKA

Illustrative composite; every step is a real function listed on the previous slide. If she is not watched, she enters her normal PIN and no alert is raised. A wrong PIN sends her guardian a heads-up; if she cannot answer, the server alerts her guardians after the window. Exporting a record and verifying it on the public ledger was run live on 26 Sep with a test member.

Built so the expensive parts don't grow per user.



Why it scales	Evidence
Sound is judged on the phone — no server GPU per member	LiteRT + YAMNet in the app
The API is stateless; state lives in PostgreSQL, so App Service can scale out	FastAPI on Azure App Service
Est. Hedera message fees ≤ R569.47/month at any member count (roots coalesced to ≤ 1 per 60 s + hourly; \$0.0008/message; R16.2075/\$)	ESTIMATE · economics script
Verification runs in each viewer's browser, not on our server	Public ledger, GitHub Pages
Only 33-byte roots go on chain; personal data never does	spec §10; D9

Layer	Technology
Phone	React Native + Kotlin, LiteRT (YAMNet), Android Keystore P-256
Server	Python FastAPI, PostgreSQL, outbox workers, Azure
Ledger	Hedera Consensus Service (testnet) via a Node sidecar
Crypto	SHA-256 chain, RFC 6962 Merkle, Ed25519, AES-256-GCM; ML-DSA-65 (demo)

Checked, not claimed.

Claim	Evidence	Tag
The server runs on Azure	/healthz 200; HTTP→HTTPS 301; TLS 1.3 negotiated (probe, 27 Sep 05:05)	FACT
Records are anchored on a public ledger	Hedera testnet topic 0.0.10687280, root #8 at 20:31:49 UTC, 26 Sep; checked in a browser: LIVE-VERIFIED	FACT
The server sends real sign-up email codes	First code delivered and verified at 02:28 UTC, 27 Sep (n = 1), from the native build; VIGIL's code step is still simulated	FACT
Wrong-PIN heads-up works end to end	Live on Azure with test phones: 5/5 (27 Sep)	FACT
Server test suite	175 passed, 2 failed (both traced to a local database shared with demo workers), 27 Sep	FACT
Browser/shared crypto library	171 tests passed (canonical JSON, DER, Merkle, verifier, post-quantum)	FACT
Anchor library	160 passed, 1 skipped	FACT
VIGIL app tests (main, 0.0.15)	jest 281 passed; TypeScript clean (27 Sep)	FACT
Every commit is scanned for secrets	Gitleaks pre-commit hook blocked a commit on 27 Sep until test secrets were hashed	FACT
Detection accuracy	Measured on public clips (right); not measured on real streets	FACT / not measured

Public test set (FSD50K eval)	n	A check-in opens
Shatter	70	72.9%
Gunshot	108	54.6%
Glass	139	46.0%
Screaming	108	6.5%

Same model and rules as the app. Screams are mostly missed: most test clips are shorter than the ~1.5 s voice rule, a trade-off chosen so a short shout doesn't cry wolf. The duress PIN and Emergency button work without any sound. Thresholds are UNCALIBRATED.

Built for a stressed person, at night, on a budget phone.

Design choice	Why
Always on after set-up; nothing to open	Coercion removes the free hand; nothing to press at the moment it matters
The check-in opens over the lock screen	No unlocking, no finding the app
The duress PIN shows the same screen as the normal PIN	Nobody watching can tell
A sound only asks; it never alerts on its own	Everyday noise costs one PIN, not a false alarm
Guardian sees who and why in words, and one big Call 10111	No interpretation needed; 10111 opens the dialer (Android requires the final press)
Calling the member is locked during an incident	A ringing phone could put them in more danger
Ivory, Silver and Midnight themes; plain wording; a visible listening notification	Readable in daylight and at night; discreet, not invisible

Can it be used tomorrow?	
Download and install	Yes — public APK, Android 7.0+
Invite a guardian and see “Guardian added”	Yes — live on Azure
Guardian alerts in the background	Yes — Android services; push (FCM) not configured
Real email code in the app's sign-up	Server yes; VIGIL's screen still simulated
Detection calibrated on real streets	Not yet

Android only for now, not iOS. VIGIL keeps the microphone open in the background during an active journey, which Android allows through a foreground service with a visible notification; iOS does not allow it at the depth needed.

Usability with real users is not yet measured; the pilot measures false check-ins per armed hour, missed events and guardian response time.

Others notice or prove. VUKA does both, in one record.

Product	Works without a press or typed code?	Leaves a record anyone can check?	Note (source)
FNB GuardMe	No — push a panic button	No	R19.90/month (FNB, Apr 2022)
Namola	No — hold for 2 seconds	No	Free panic app
Discovery Bank Panic Code	No — type a code	No — inside one bank	MyBroadband, 27 Mar 2025
Google Personal Safety · Apple Check In	Partly (crash detection / missed arrival)	No	Detect, no proof
iTOO [My]Cylution	Insurance, not detection	No	From R22.50/month; covers express kidnapping (iTOO, 24 Sep 2026)
OpenTimestamps · Truepic / C2PA	No — they cannot notice	Yes	Prove, don't detect
VUKA	Yes — a sound opens the check; a second PIN asks for help	Yes — signed, chained, anchored on Hedera	Partner pays ~R20/member/month (ASSUMPTION)

What they do better: GuardMe and Namola connect to armed response today; banks' own duress codes act inside the bank instantly. VUKA is not a response service and never dispatches anyone — it notices, alerts a person, and keeps the proof.

Imminent risks, and what the design does about them.

Threat	What the design does	Status
Attacker holds the unlocked phone and ignores the check-in	The server escalates on no answer	Built
Attacker watches the duress PIN being entered	Same screen and flow for both PINs	Built; timing-parity test open
Attacker forces the normal PIN	Records “normal code entered”; a normal PIN alone never closes an incident	Designed (threat model)
Staged fraud (an accomplice screams)	The anchor proves when, not what; a missing record is never used against anyone	Stated limit
Compromised server rewrites history	Chain + public roots let a stranger detect rewriting	Built; suppression is a stated limit
Leaked keys or secrets	Secret scan on every commit and in CI; old keys rotated, history purged	Built
Replay of a signed request	Single-use nonce, ±120 s clock window	Built

Secure lifecycle (NIST SSDF)	Proof
Prepare: STRIDE + coercion threat model; 79 tracked controls	Scorecard (25 Sep): 16 done · 30 in build · 29 planned · 4 not doing, with reasons
Protect: Gitleaks (checksum-pinned) on commits and full history	checks.yml; blocked a commit on 27 Sep
Produce: Semgrep, OSV-Scanner, npm + pip audit, contract tests	checks.yml
Verify: public anchor; LIVE-VERIFIED root #8	Hedera mirror

Ethics, by construction: the machine may decide what it is hearing, never what happens to a person; guardians get the alert, not the record; no audio is ever kept; we never claim “unhackable”, “court-admissible” or “prevents crime”. No independent penetration test yet.

Where each control sits (OSI and TCP/IP).

OSI layer	TCP/IP layer	VUKA control	Status
7 Application	Application	Every event signed on the phone (P-256) and every server entry signed (Ed25519); signed requests; PIN authorisation for sensitive actions; duress parity	Built
6 Presentation	Application	Canonical JSON (spec §5) so a signature covers exact bytes; payloads encrypted at rest (AES-256-GCM); hybrid ML-DSA-65 attestation	Built; PQ is a demo
5 Session	Application	No sessions or cookies: each request carries its own signature, a single-use nonce (kept 24 h) and a ± 120 s timestamp window	Built
4 Transport	Transport	TLS 1.3 to the Azure endpoint; HTTP redirected to HTTPS (301). Post-quantum key exchange in TLS depends on Azure and Android	Observed 27 Sep
3 Network	Internet	Azure App Service public endpoint; database firewall allows only Azure services; Hedera public network for anchors	Configured (Sibusiso, 26 Sep)
2 Data link · 1 Physical	Link	Keys stay in Android Keystore (StrongBox when present); raw audio never leaves the phone's 3-second buffer; microphone indicator stays visible	Built

Why this matters: the attacker in our threat model is often physically present — so the most important controls live at the top (what the screen and the record reveal) and at the bottom (what never leaves the phone).

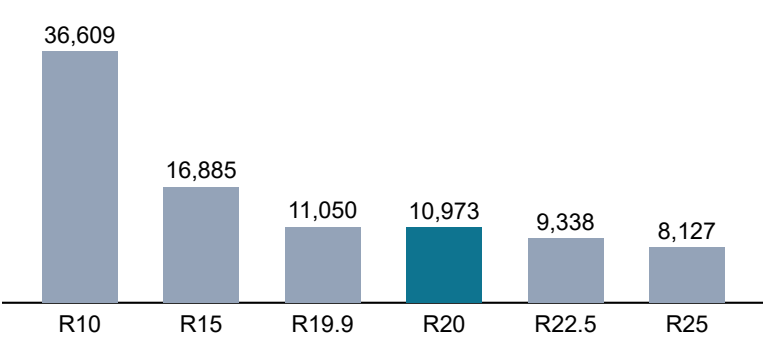
Collect little, keep it short, publish nothing personal.

Data	Why	Where	How long	Who can see it
Sound	To notice danger	Only on the phone: a label and score; raw audio is never stored (3-second buffer)	Seconds	No one
Signed events (check-ins, alerts)	The member's record	Server, payloads encrypted (AES-256-GCM)	90 days specified; not yet verified as implemented	The member; guardians see alerts only
Location	Help a guardian reach her	Sent only after a check-in; kept only while a guardian alert is open	Purged 24 h after the alert ends (PROPOSED)	Alerted guardians
Account (email, phone, name)	Sign-up and recovery	Server, encrypted; passwords and codes hashed (Argon2id)	While the account exists	The member
On the public ledger	Proof of when	Hedera: 33-byte roots only	Permanent	Anyone — nothing personal in it

Your rights and our basis	
Lawful basis	Consent, given in the app before use
Access	The member can see and export her record
Deletion	Payload removed, hash kept (so no one can pretend it never existed); 72-hour cooling-off specified
Guardians	Consent at enrolment (POPIA s18 notice); can leave at any time
Open, stated	No legal entity yet; Information Officer registration prepared, not submitted; counsel questions recorded

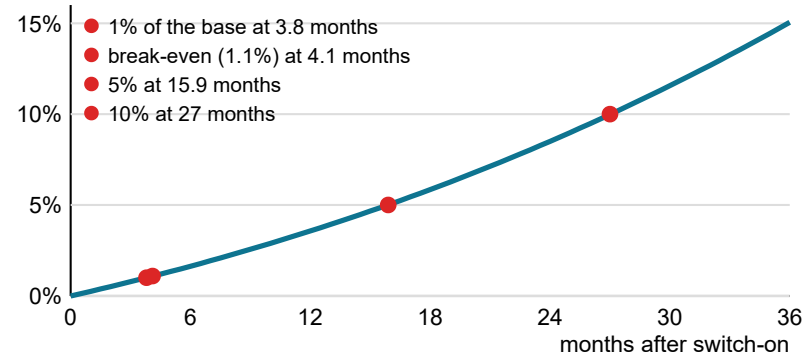
The victim pays nothing. The insurer and the bank pay because it saves them more.

Members to break even, by monthly price (5 stipends)

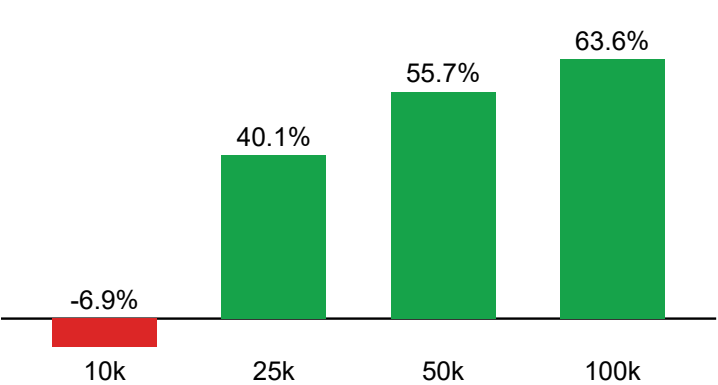


scripts/economics_vigil_anchor.py · ESTIMATE · R20 is the decided price

Adoption in one 1M-customer channel (reference curve)



Operating margin at R20, by members



same script · ESTIMATE

Who pays	
Member	R0
Insurer	R20 per member per month · ASSUMPTION
Bank	R15 per case · PROPOSED

Go-to-market	What happens	Proof point
1 · Pilot (12 months)	One insurer, 5,000 members; one bank to cost a dispute	Actuary test: insurer saves ≥ R240/member/year
2 · Switch-on	Partner turns VUKA on inside its app for its customers	Break-even ~1.1% of a 1M base, ~4.1 months (reference, not forecast)
3 · Scale	More insurers and banks; per-case fee on disputes	Fixed ledger fees at any member count

Unit economics	
Priced where people already pay	GuardMe R19.90 · VUKA R20 · iTOO R22.50 (FACT)
Variable cost per member	R5.72/month · contribution 71.4% (ESTIMATE)
Fixed cost, 5 stipends	R156,683.98/month; product alone R6,683.98 → 469 members (ESTIMATE)
First-year funding ask	R1,884,583: team R1.8m + product/ledger R80,918 + one-off R3,665 (ESTIMATE)

Ready for a future that can forge today's signatures.

Why it matters to VUKA: a record may be checked years later, in a dispute or a court. A large quantum computer running Shor's algorithm could forge the elliptic-curve signatures used today (Ed25519, P-256), and anything protected by an elliptic-curve key exchange can be recorded now and decrypted later. The anchored chain itself is SHA-256 and Merkle trees (Grover leaves ~128-bit security), so committed bytes and their timing stay checkable.

Algorithm (NIST standard)	Use in VUKA	Status 27 Sep	Time to production (ESTIMATE)
ML-DSA-65 (FIPS 204) + Ed25519, both required	Sign each anchored root; the verifier also checks the Hedera mirror record (topic, sequence, time, exact 33 bytes)	Implemented and tested as a demonstration on the real testnet root #8; test keys, not yet pinned (PR #117)	2–3 days: key ceremony, pin the key in the manifest, sign every new root, verify in the public ledger
ML-KEM-768 (FIPS 203)	Protect a record shared with a bank or insurer against harvest-now-decrypt-later	Interoperability proven between the server and browser libraries; the hybrid sealed export (X-Wing) is designed, not built	1–2 weeks, with published test vectors and recipient-key authentication
SLH-DSA (FIPS 205)	Hash-based signer for the long-lived key manifest	Outlined	3–5 days once a reviewed signer is chosen
Post-quantum TLS; phone keys	Transport; device identity	Depends on Azure and on Android Keystore support	Platform-dependent

Evidence: two independent implementations — pyca/cryptography 50 on the server and @noble/post-quantum in the browser — derive byte-identical ML-DSA-65 and ML-KEM-768 keys from the same seed, verify each other's signatures and decapsulate each other's ciphertexts; 28 Python and 171 JavaScript tests pass, including tampering cases. The plan and the code were each reviewed adversarially before submission. **Not claimed:** “quantum-resistant” or “post-quantum ready”; @noble/post-quantum states it has not been independently audited.

We are not asking you to believe us. We are asking you to check us.

Who	Impact
The person under threat	Help without a free hand; a record made at the time instead of her word against a payment
Her guardian	Told silently, with the reason in words and one button to 10111
Insurers	A way to tell a recorded coercion from a story — the pilot measures fewer fake claims paid and faster settlement
Banks	Disputes settled on evidence instead of escalated (~R4,962 and ~61 days per ombud case)
What the pilot must measure before any saving is claimed	
False check-ins per armed hour · missed events · alerts delivered · guardian response time	
Evidence packs pulled · claims-handling time with and without a record · fraud leakage	

Limits, stated first	
Detection	Thresholds uncalibrated; screams mostly missed on public clips; no street recordings yet
App	Push (FCM) not configured in this build; VIGIL's sign-up code step is simulated; always-on listening and graded check-ins are PROPOSED designs
Ledger	Hedera testnet, not mainnet; the anchor proves when, not what
Security	No independent penetration test; the key rotation awaits security review
Legal	No legal entity yet; Information Officer registration prepared, not submitted
Market	The number of coerced transfers is unknown; prices are an assumption until a partner's data tests them
Check it yourself	
Public ledger	https://lethabomh14.github.io/V.U.KA--Geekulcha/dashboard/ledger/
Code	https://github.com/LethaboMH14/V.U.KA--Geekulcha